Mikor mondhatjuk, hogy a programunk holtpontban van?

VIZSGÁLAT

$A_i \to$ külső await-el (nem párhuzamos blokkban vannak)

$T_j \to$ párhuzamos blokkok

$\vdots$

A_1: await β_1 then S_1 ta

$\vdots$

A_i: await β_i then S_i ta

$\vdots$

T_j: parbegin $S_1^j \| S_2^j \| \dots \| S_{n_j}^j$ parend

n ... külső await-el száma

m ... párhuzamos blokkok száma

$j \in [1..m]$ esetén n_j ...

T_j programkomponensinek száma

$$D(S) = \left[\bigvee_{i=1}^{n} (pre(A_i) \wedge \neg\beta_i)\right] \vee \left[\bigvee_{j=1}^{m} D1(T_j)\right], \text{ ahol}$$

logikai
függvény
$A - n$

$$D1(T_j) = \left[\bigwedge_{k=1}^{n_j} \left(post(S_k^j) \wedge D(S_k^j)\right)\right] \wedge \left[\bigvee_{k=1}^{n_j} D(S_k^j)\right]$$

$\forall a \in A : D(S)(a) = \text{hamis}$

$D(S)$ azonosan hamis.

Ha $D(S)$ azonosan hamis, akkor S biztosítja a holtpontmentes.

$i, j := 1, 1$

$\{a = a' \wedge i = 1 \wedge j = 1\}$

parbegin $S_1 \| S_2$ parend

$\}\ S$

S_1:

$\{Inv\}$

while $i \leq n$ do

$\{Inv \wedge i \leq n\}$

await $i = j$ then

$x, i := a[i], i+1$

ta

$\{Inv\}$

od

$\{Inv \wedge i = n+1\}$

$\{post(S_1)\}$

S_2:

$\{Inv\}$

while $j \leq n$ do

$\{Inv \wedge j \leq n\}$

await $i > j$ then

$b[j], j := x, j+1$

ta

$\{Inv\}$

od

$\{Inv \wedge j = n+1\}$

$\{post(S_2)\}$

$$D(S) = \left[\bigvee_{i=1}^{n} (pre(A_i) \wedge \neg \beta_i) \right] \vee \left[\bigvee_{j=1}^{m} D1(T_j) \right]$$

$$\boxed{\begin{array}{l} \{pre(A_i)\} \\ A_i \\ \{post(A_i)\} \end{array}}$$

$n = 0$

$$\bigvee_{i=1}^{0} pre(A_i) \wedge \neg \beta_i = HAMLS$$

$m = 1$

$$\bigvee_{j=1}^{1} D1(T_j) = D1(T1) \overset{n_1=2}{=\!=} \left[\bigwedge_{K=1}^{2} (post(S_K) \vee D(S_K)) \right] \wedge \left[\bigvee_{K=1}^{2} D(S_K) \right] =$$

$$= (post(S_1) \vee D(S_1))$$

É'S

$$(post(S_2) \vee D(S_2))$$

É'S

$$(D(S_1) \vee D(S_2))$$

helyett

$$post(S_1) \wedge D(S_2)$$

VAGY

$$post(S_2) \wedge D(S_1)$$

VAGY

$$D(S_1) \cap D(S_2)$$

$$post(S_1) \wedge D(S_2) = HAMIS$$

$$(Inv \wedge i = n+1) \wedge (Inv \wedge j \leq n \wedge i \leq j) = HAMIS$$

$$\underline{n+1} = i \leq j \underline{\leq n} \quad \lightning$$

$$post(S_2) \wedge D(S_1) = HAMIS$$

$$(Inv \wedge j = n+1) \wedge (Inv \wedge i \leq n \wedge i \neq j)$$

$$\underline{n+1} = j \underset{(Inv)}{\leq} i \leq n \quad \lightning$$

$D(S_1) \wedge D(S_2) = \text{HAMIS}$

$(\text{Inv} \wedge i \leq n \wedge i \neq j) \wedge (\text{Inv} \wedge j \leq n \wedge i \leq j) = \text{HAMIS}$

$$\frac{\begin{array}{c} j \leq i \\ i \leq j \end{array}}{\begin{array}{c} i = j \\ i \neq j \end{array}} \lightning$$

$D(S)$ azonosan hamis.

Tehát S holtpontmentes.

$x : \mathbb{Z}$

$\{x=0\}$

parbegin $S_1 \| S_2$ parend

$\{x=1\}$

S

S_1:

$\{x=0 \vee x=1\}$

await $x=1$ then skip ta

$\{x=1\}$ — $\{post(S_1)\}$

S_2:

$\{x=0\}$

$x := 1$

$\{x=1\}$ — $\{post(S_2)\}$

$$D(S) = \left[\bigvee_{i=1}^{n} (pre(A_i) \wedge \neg \beta_i)\right] \vee \left[\bigvee_{j=1}^{m} DA(T_j)\right] \qquad \underline{\underline{\begin{matrix} n=0 \\ m=1 \end{matrix}}}$$

$$\bigvee_{j=1}^{1} DA(T_j) = DA(T_1) \overset{n_1=2}{=} \left[\bigwedge_{k=1}^{2} post(S_k) \vee D(S_k)\right] \wedge \left[\bigvee_{k=1}^{2} D(S_k)\right] =$$

$$= \left(post(S_1) \vee D(S_1)\right) \wedge \left(post(S_2) \vee D(S_2)\right) \wedge \left(D(S_1) \vee D(S_2)\right)$$

$$\left(post(S_1) \wedge D(S_2)\right) \vee \left(post(S_2) \wedge D(S_1)\right) \vee \left(D(S_1) \wedge D(S_2)\right)$$

$post(S_1) \wedge D(S_2)$

$(x = 1) \wedge HAMIS$

$post(S_2) \wedge D(S_1)$

$(x = 1) \wedge ((x = 0 \vee x = 1) \wedge x \neq 1)$

$x = 1$
$x \neq 1$

$= HAMIS$

$D(S) = HAMIS$, wirt

S holtpontmontes.

$D(S_1) \wedge D(S_2)$

$((x = 0 \vee x = 1) \wedge x \neq 1) \wedge HAMIS$

$A = (x : 0)$

S_1:

$\{x = 0 \lor x = 3\}$

$x := x + 2$

$\{x = 2 \lor x = 5\}$

S_2:

$\{x = 0 \lor x = 2\}$

$x := x + 3$

$\{x = 3 \lor x = 5\}$

Mutassuk meg, hogy a $\underbrace{\text{parbeli } S_1 \| S_2 \text{ parend}}_{\text{rinsbenne await, tehát holtpontmentes}}$ holtpontmentes.

$\{x = 0\}$

$z := 1$

$\{Inv\}$

parbegin $S_1 \| S_2$ parend

$\{z = x^{'n'}\}$

S

S_1:

$\{Inv\}$

while $n \neq 0$ do

$\{Inv \wedge n \neq 0\}$

$n, z := n-1, z \cdot x$

od

$\{z = x^{'n'} \wedge n = 0\}$

S_2:

$\{Inv\}$

while $n \neq 0$ do

$\{Inv\}$

await even(n) then

$x, n := x \cdot x, n/2$

ta

od

$\{z = x^{'n'} \wedge n = 0\}$

Hoffpontmentessig

$$D(S) = \left[\bigvee_{i=1}^{n} (pre(A_i) \wedge \neg p_i)\right] \vee$$

$n = 0$

$$\left[\bigvee_{j=1}^{m} DA(T_j)\right]$$

$m = 1$

$$\bigvee_{j=1}^{1} DA(T_j) = DA(T_1)$$

$n_1 = 2$

$$\left[\bigwedge_{K=1}^{2} (post(S_K) \vee D(S_K))\right] \wedge \left[\bigvee_{K=1}^{2} D(S_K)\right]$$

$$(post(S_1) \vee D(S_1)) \wedge (post(S_2) \vee D(S_2)) \wedge (D(S_1) \vee D(S_2))$$

helyett

$post(S_1) \wedge D(S_2)$ VAGY

$post(S_2) \wedge D(S_1)$ VAGY

$D(S_1) \wedge D(S_2)$ mikor azonosan hamis?

$post(S_1) \wedge D(S_2)$

$(z = x^{'n'} \wedge n = 0) \wedge (Inv \wedge odd(n))$ azonosan hamis

even(0), odd(0)

$post(S_2) \wedge D(S_1)$

$(z = x'^{n'} \wedge n = 0) \wedge HAMIS$ azonosan hamis.

$D(S_1) \wedge D(S_2)$

$HAMIS \wedge (Inv \wedge odd(n))$ azonosan hamis.

$D(S)$ azonosan hamis, tehát

Shottportmentes.